CASEY DEL CARPIO BARTON

APPLIED AI SYSTEMS ARCHITECT | Agent Infrastructure Engineer

Forward-Deployed AI | Builds the operating layer that makes powerful AI dependable enough to use.

Honolulu, Hawaii
808-936-5654
glacier.equilibrium@gmail.com
casey-barton-glaciereq.vercel.app
github.com/GlacierEQ

SYSTEMS SIGNAL

Applied AI systems architect who turns ambitious, ambiguous ideas into bounded operating systems with explicit authority, deterministic evidence, controlled failure behavior, and inspectable completion receipts. The method is simple: observe the actual system, separate evidence from inference, isolate the governing failure, build the narrowest useful mechanism, test the stated behavior, expose the limit, and leave a usable artifact.

69/69	166 + 19	67	95/95
Receipt Router tests	source + memory tests	Helix admitted public set	Agent Coordinator tests

PROOF AT WORK

Professional Experience

GlacierEQ | Founder / Applied AI Systems Builder

2023-Present

- Design and implement evidence-bound systems spanning agent orchestration, application intelligence, document and evidence pipelines, memory and continuity, infrastructure governance, and human-machine interfaces.
- Convert broad architecture into typed contracts, runnable slices, deterministic tests, explicit refusal behavior, machine-readable facts, and reviewable completion receipts.
- Built a public Portfolio Receipt Router that replaced unsupported autonomous-control semantics with a local fail-closed evidence system; 69/69 tests passed through direct and reusable CI with zero external actions.
- Govern a multi-repository engineering estate by separating source presence, review, execution, deployment, authority, and verified completion so public claims cannot outrun proof.

Diamond Head Home Inspections | Certified Home Inspector

2020-2024

- Inspected residential structural, roofing, electrical, plumbing, HVAC, moisture, and safety conditions; translated field evidence into clear, decision-ready reports under time constraints.
- Built disciplined habits around uncertainty labeling, defect prioritization, interacting-system analysis, client communication, and avoiding unsupported conclusions.

Hi-Class Home Services / Hi Class Maintenance Oahu LLC | Owner-Operator, Building Systems & Field Services

2017-Present

- Scope residential repair and service work, prepare estimates, coordinate execution, and communicate assumptions, constraints, and completion status with clients.
- Translate incomplete real-world requirements into bounded work packages, explicit assumptions, and usable closeout artifacts.

CAPABILITY FIELD

Core Capabilities

Agent infrastructure & governance

Multi-agent coordination; MCP and tool contracts; capability/authority separation; approval models; state machines; bounded retries; recovery and closure.

Evidence & release engineering

Provenance; claim-to-source mapping; deterministic testing; JSON Schema; CI/CD; static analysis; dependency auditing; integrity hashing; machine-readable receipts.

Application & document intelligence

Role research; evidence-qualified resume generation; document extraction; structured artifacts; human and machine presentation surfaces.

Primary technology

Python; TypeScript; JavaScript; SQL; Bash; Node.js; React; Next.js; FastAPI; REST; JSON-RPC 2.0; MCP; GitHub Actions; Docker; Vercel.

SELECTED SYSTEMS

Evidence states, not hype

Portfolio Receipt Router TEST VERIFIED Fail-closed evidence routing and compatibility-safe architecture repair.	EVIDENCE 69/69 tests; direct and reusable CI; artifact 8910423397; zero external actions. BOUNDARY Independent technical exhibit; no production-operation or company-affiliation claim.
Job Application Helix PARTIALLY VERIFIED Evidence-governed hiring and portfolio orchestration with an exact 67-repository admitted public proof boundary.	EVIDENCE Fail-closed evidence levels, repository-state separation, promotion gates, machine contracts, and role-calibrated package generation. BOUNDARY The 67 admitted repositories are a public proof set, not accomplishment count or the full owned estate; child repositories retain independent evidence states.
Agent Coordinator RECORDED 95/95 Deterministic task ownership, dependency order, capacity, stable priority, and shared budgets.	EVIDENCE 95/95 recorded Python tests. BOUNDARY Hosted matrix and broader promotion gates remain separate.
AKOS / PSYSOC-X BOUNDED VERIFIED SURFACES Authority, evidence, closure, and deterministic audience-calibration primitives.	EVIDENCE Repository-native verification supports bounded current surfaces. BOUNDARY Presentation depth can change; facts, uncertainty, dignity, and authority may not.

SYSTEMS FOUNDATION

Cross-domain operating perspective

Scientific measurement	University of Hawaii sea-urchin morphometric research, 2016-2017: processed physical measurements of gill structures to support visual species identification.
Compressed-gas systems	Scuba Tank Technician, Hawaiian Diving Adventures and UH Dive Office, 2016-2017: tank filling, transport, inspection and repair support, and gas mixing under safety-critical procedures.
Field systems operations	Residential inspection and repair work developed practical judgment around interacting systems, failure consequences, uncertainty, and clear client communication.

TECHNICAL PROFILE

Tools and architecture

Primary: Python, TypeScript, JavaScript, SQL, Bash, Node.js, React, Next.js, FastAPI

Architecture: REST, JSON-RPC 2.0, MCP, JSON Schema, Protocol Buffers, events, state machines, idempotency, authority models, evidence models

Delivery: Git, GitHub Actions, Docker, Vercel, automated testing, static analysis, dependency auditing, deterministic release checks

Additional evidence-bounded exposure: Go, Rust, C, C++, CUDA, Swift/Metal, Julia, R, Verilog/SystemVerilog, WebAssembly

EDUCATION

Scientific foundation and current cloud training

University of Hawaii at Manoa — B.S., Marine Biology - 2016

AWS Cloud Institute — Cloud Application Developer program - 2025-2026, in progress

EVIDENCE BOUNDARY Independent GlacierEQ work and bounded technical exhibits. Test counts refer only to their stated repository and scope. No company affiliation, proprietary access, production deployment, customer impact, formal people-management experience, current certification status, hardware validation, or measured business outcome is claimed without direct evidence.